

Question 1. Estimate the error if $P_3(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6}$ is used to estimate the value of $f(x) = e^x$ by filling in the blanks below.

1. By Taylor's theorem, for each x , the error is

$$R_3(x) = f(x) - P_3(x) = \frac{f^{(4)}(c)}{4!} x^4 = \underline{\hspace{2cm}} x^4$$

for some c in between 0 and x .

2. Noting that $|c| \leq |x|$, the error is bounded by (fill in a number)

$$|R_3(x)| \leq \underline{\hspace{1cm}} e^{|x|} |x|^4.$$

3. If we were to consider all x in $(-1, 1)$, an upper bound for the error is (fill in a number)

$$|R_3(x)| \leq \underline{\hspace{1cm}}$$

4. In general, if we were to consider all x in $(-a, a)$, an upper bound for the error is

$$|R_3(x)| \leq \underline{\hspace{2cm}}$$

depending on the value of a .

If we want to find an interval $(-\frac{1}{N}, \frac{1}{N})$ such that the error is less than 0.001, we need to solve the inequality

$$|R_3(x)| \leq \underline{\hspace{2cm}} < 0.001,$$

so one possible value of N is $\underline{\hspace{1cm}}$.

Question 2. Calculate an upper bound for the error if the following $P_n(x)$ is used to estimate $f(x)$ on $(-0.1, 0.1)$. Moreover, find an interval $(-a, a)$ such that the error is less than 0.001.

a) $P_1(x) = 1 + \frac{x}{2}$ for $f(x) = \sqrt{1+x}$

b) $P_4(x) = x - \frac{x^3}{6}$ for $f(x) = \sin(x)$

(this P_4 is indeed a fourth-degree approximation of $\sin(x)$)

c) $P_2(x) = x + x^2$ for $f(x) = \sin(x)e^x$

d) $P_3(x) = \frac{x^3}{3}$ for $f(x) = \int_0^x \sin(t^2) dt$